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Collective prompt tuning with relation inference for document-level relation extraction

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ABSTRACT

Document-level relation extraction (RE) aims to extract the relation of entities that may be across sentences. Existing methods mainly rely on two types of techniques: Pre-trained language models (PLMs) and reasoning skills. Although various reasoning methods have been proposed, how to elicit learnt factual knowledge from PLMs for better reasoning ability has not yet been explored. In this paper, we propose a novel **C**ollective **P**rompt **T**uning with **R**elation **I**nference (CPT-RI) for Document-level RE, that improves upon existing models from two aspects. First, considering the long input and various templates, we adopt a collective prompt tuning method, which is an update-and-reuse strategy. A generic prompt is first encoded and then updated with exact entity pairs for relation-specific prompts. Second, we introduce a relation inference module to conduct global reasoning overall relation prompts via constrained semantic segmentation. Extensive experiments on two publicly available benchmark datasets demonstrate the effectiveness of our proposed CPT-RI as compared to the baseline model (ATLOP (Zhou et al., 2021)), which improve the 0.57% on the DocRED dataset, 2.20% on the CDR dataset, and 2.30 on the GDA dataset in the F1 score. In addition, further ablation studies also verify the effects of the collective prompt tuning and relation inference.

1. Introduction

Relation extraction (RE) is a fundamental task in the field of Information Extraction (IE). It aims at identifying relations between two given entities from texts, where the texts may be at the sentence level (Miwa & Bansal, 2016; Santos, Xiang, & Zhou, 2015), at the document level (Li et al., 2016; Yao et al., 2019), or at the corpus level (Yao et al., 2021). The extracted relations reveal the underlying structures of texts and benefit a wide spread of applications, such as question answering (Chen, Chang, Chen, Nayak, & Ku, 2019; Yu et al., 2017) and knowledge graph construction (Chen et al., 2022; Li, Sun, Zhang and Zhang, 2021).

Following the success of sentence-level RE, several document RE benchmarks (Zeng, Xu, Chang, & Li, 2020; Zhang, Chen et al., 2021) are established to promote models in dealing with additional challenges of cross-sentence reasoning. As shown in Fig. 1, although `Ralph` and `United States` are located at different sentences, we can still know that they have a relation `country of citizenship`. Recent research attention is to develop various forms of reasoning methods to capture or bypass long-distance dependencies, including building an entity/mention graph (Nan, Guo, Sekulic, & Lu, 2020; Zeng et al., 2020) or relation matrix (Li, Xu et al., 2021; Zhang, Chen et al., 2021), and modeling logic rules (Ru et al., 2021) or paths (Huang et al., 2021; Xu, Chen, & Zhao, 2021). However, the above reasoning module is built on top of Pre-trained Language Models (PLMs), thus the reasoning quality heavily depends on the contextualized representations from PLMs.

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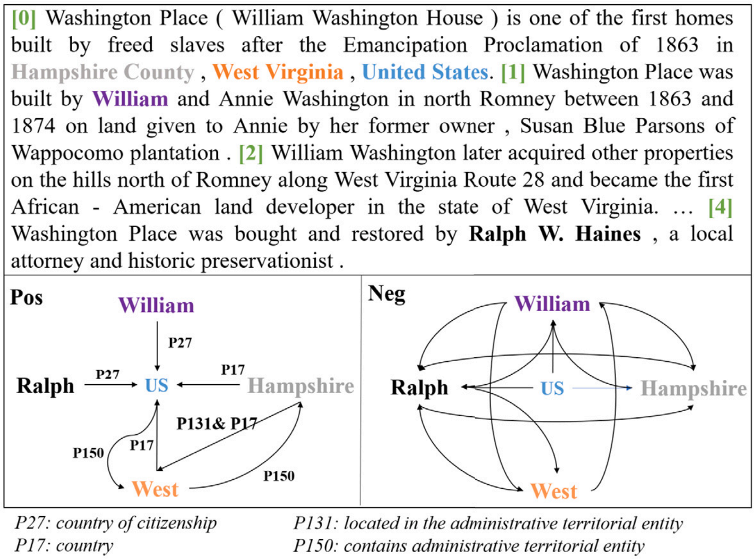

Fig. 1. Example of document-level RE, where Pos denotes positive relations, and Neg denotes no relation between two entities.

In this paper, we propose to better elicit factual knowledge from PLMs via prompt tuning, and then empower with reasoning ability for document RE. Prompt tuning is a new paradigm of NLP to efficiently and effectively leverage PLMs (Liu, Yuan et al., 2021; Nie et al., 2022) for downstream tasks. It usually designs specific input formats either using additional soft prompt vectors or natural language templates, so that the output becomes more similar to original objectives during pre-training. However, prompt tuning techniques are hardly adapted into tasks with complex output spaces. The key challenge here is that complex settings will greatly increase the difficulty in prompt design. Taking document-level RE as an example, a document usually contains tens of entities. To predict relations between entities, each pair of entities may require different prompt templates. How can we manually design and tune them? Not to mention jointly reasoning over all entity pairs may lead to overlong inputs (far exceeding the limits of current PLMs, e.g., 512 in BERT Devlin, Chang, Lee, & Toutanova, 2019).

Another major issue caused by exhaustive entity pairs is dominated false positives. Most entity pairs have no relations, bringing noisy interactions for jointly reasoning. As shown in Fig. 1, there are many more negative relations than positives in the document. In fact, this contains 342 entity pairs with 4.92% positive relations only.

To address the issues, we propose a novel **C**ollective **P**rompt **T**uning with **R**elation **I**nference method for Document-level RE, namely CPT-RI. The basic idea is to seamlessly integrate two modules of collective prompt tuning and relation inference. For collective prompt tuning, we aim at efficiently leveraging many prompts for all possible entity pairs. In specific, there are two steps. First, we design a generic template including both entity and relation placeholders. Then, the encoded generic prompts will be re-organized and updated for each relation-specific prompts. Thus, each possible entity pair will obtain its own prompt-guided representations.

For relation inference, we aim at capturing high-order relations among entity pairs for knowledge transfer, while depressing the impacts of noisy false positives. Inspired by semantic segmentation (Zhang, Chen et al., 2021), we build a relation matrix based on collective prompt tuning, where each element in the matrix denotes the relation feature of entity pair from the above module. By segmenting different groups, relations are encouraged to combine with other informative relations to infer new relations. Besides, we also introduce criss-cross constraints over the matrix, so that only relations with overlapped entities shall be visible to each other, to mitigate unnecessary interactions.

The contributions of this paper can be summarized as follows:

- We propose a novel collective prompt tuning method that can efficiently and effectively deal with complex and overlong input templates.
- We use relation inference to mitigate confounding interactions for relations and enhance relation reasoning.
- We conduct extensive experiments on three publicly available datasets. Compared with 22 baseline models, the results show a superior performance of our proposed CPT-RI model, and further ablation study demonstrates the effectiveness of the key components.

The rest of this article is organized as follows. Section 2 describes the related work about document-level RE and prompt tuning. Section 3 is the Research Objective which aims to highlight our motivation and methods. In Section 4, we briefly describe the preliminaries and framework of our model. In Sections 5 and 6, the Collective Prompt Tuning and Relation Inference are proposed and presented its inference process. In Section 7, we report the results of the experiment and conduct the various ablation studies. Finally, the conclusion of this work is made in Section 8.

2. Related work

There are two related research areas: document-level RE and prompt tuning.

2.1. Document-level relation extraction

In recent years, document-level natural language processing tasks have been favored by many researchers, such as document-level event extraction (Zhao, Ji, He, Liu, & Ren, 2021), document-level named entity recognition (Lopez, Du, Cohoon, Ram, & Howison, 2021), and document-level RE (Li, Xu et al., 2021; Liu, Tan and Dong, 2021). The task of RE is to identify relational facts between entities from plain text. Traditional RE (Shi et al., 2018; Shi, Feng, Xu, Liao, & Huang, 2020; Wen, Liu, Ouyang, Lin, & Chung, 2021; Ye & Luo, 2020) methods focus on sentence-level RE due to the lack of document-level RE datasets. Recently, Yao et al. (2019) proposes a new dataset about document-level RE and some baselines. Existing works mainly rely on two types of techniques: enhancing context information and logical reasoning through various graphs. To enhance the context information, Wang, Hu, Cao, and Sun (2020) is a novel model for document-level RE by encoding the document information in terms of entity global and local representations as well as context relation representations. Zhou, Huang, Ma, and Huang (2021) proposes adaptive-thresholding loss and localized context pooling to learn an adaptive threshold and transfer attention to locate relevant context, which adaptive-thresholding loss has become the most classical method for document-level RE. To capture the relation reasoning among the entity pairs, Nan et al. (2020) empowers relational reasoning across sentences and develops a refinement strategy, which enables the model to aggregate relevant information incrementally. Zeng et al. (2020) constructs a heterogeneous mention-level graph (hMG) to model complex interaction and constructs an entity-level graph (EG) to infer relations between entities. Makino, Miwa, and Sasaki (2021) treats the relation as a relation graph that can classify them in a close-first manner using the document and temporally constructed graph information. Zhang, Wong et al. (2021) proposes to incorporate explicit discourse modeling and leverage modular self-supervision for each sub-problem. Tran, Nguyen, and Nguyen (2020) proposes to compute the representations for the nodes in the graph-based edge-oriented model for document-level RE. Jia, Wong, and Poon (2019) combines representations learned over various text spans throughout the document and across the sub-relation hierarchy. But these methods are more focused on constructing entities/mentions graph to reason relations. To make more explicit reasoning relational information, Tan, He, Bing, and Ng (2022) and Zhang, Chen et al. (2021) employ the context information and logical reasoning to construct the graph and enhance the performance of document-level. Xie, Shen, Li, Mao, and Han (2022) propose an evidence enhanced framework to empower document-level RE by extracting evidence and fusing the extracted evidence in inference. So the significant difference between our model and previous models is that we build a novel reason graph for each entity pair to restrict unnecessary interactions.

2.2. Prompt tuning

Recently, many large-scale PLMs have emerged and become a crucial technique in both academia and industry. To alleviate the cost of fine-tuning, prompt tuning has been proposed (Brown et al., 2020). It demonstrates that simply modification of inputs can achieve significant performance gains, because a well-designed input format can convert downstream tasks to cloze style that is similar to the pre-training objectives of PLMs. Clearly, how to design a suitable prompt is the key problem. Gao, Fisch, and Chen (2021) proposes automating prompt generation and refining strategy for selectively incorporating demonstrations into each context. Hu et al. (2022) proposes to incorporate external knowledge into the verbalizer, forming a knowledgeable prompt tuning (KPT), to improve and stabilize prompt tuning. Han, Zhao, Ding, Liu, and Sun (2021) proposes prompt tuning with rules (PTR) for many-class text classification, and apply logic rules to construct prompts with several sub-prompts. Schick and Schütze (2021) proposes to utilize prompt tuning to predict multiple `[Mask]`. In this paper, we focus on how to apply prompt tuning in an efficient way to share and reuse prompts, for the tasks with complex output space, i.e., document RE with joint predictions on multiple multi-label classification.

This paper is the extension of ATLOP (Zhou et al., 2021). Compared to original work, this paper has several improvements:

Methods: (1) Collective Prompt Tuning. We propose a novel collective prompt tuning method to capture the potential language information and solve the problem of overlong and complex templates for document-level RE. (2) Relation Inference. We use Relation Inference to capture high-order relations among entity pairs and reduce the influence of noisy false positives.

Experiments: (1) We further do experiments to analyze the effectiveness of our novel Collective Prompt Tuning module. (2) The evaluation experiments on the effectiveness of Relation Inference have further been conducted. (3) Compared with ATLOP, our method has achieved great improvement.

Content: We analyze the current urgent problems of document-level RE from the perspective of pre-trained language models and relation reasoning and provide solutions, which are Collective Prompt Tuning module and Relation Inference module.

3. Research objective

This paper aims to address the following shortcomings of the existing methods for document-level relation extraction: (1) Over-length Prompt Templates. Generally, a document contains many entities, and different pairs of entities should have different prompt templates. Thus, a typical prompt-based method may have overlong inputs, leading to a degradation of performance or even inapplicable cases. (2) Massive Confounding Interactions. Although various forms of the graph have been developed to model the interactions among mentions/entities for global reasoning, there are lots of noisy interactions.

To solve these problems, we propose a novel CPT-RI. First, to address the first problem, we design a generic prompt template, then update and reuse it for all of the relations in the document. To address the second problem, we introduce criss-cross constraints into the entity matrix for global reasoning, where each element in the matrix denotes the relation between corresponding entities.

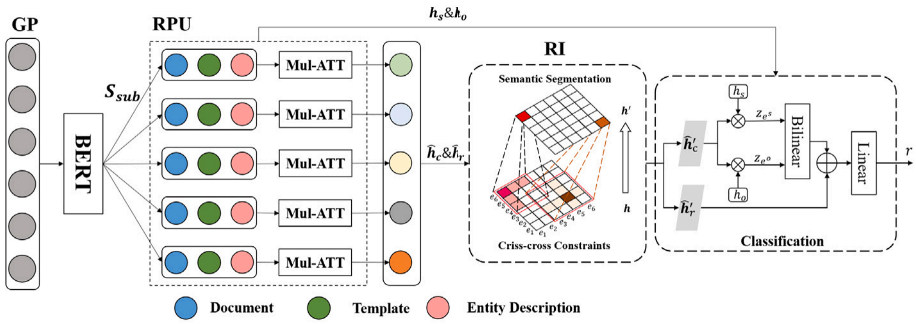

Fig. 2. Architecture of proposed model.

4. Preliminaries and framework

4.1. Preliminaries

Document RE can be regarded as a multiple multi-label classification problem. Given a sequence of document $D = [c_1, c_2, \dots, c_n]$ and contains N' entities, where n is the length of document, and these entities can form N $(N = N' * (N' - 1))$ entity pairs. Each c is a word in the document. Document RE thus aims to predict relations for each of these entity pairs. And the entity, e, may contain a number of mentions that are the same meaning words in different positions, $e = [m_1, \dots, m_k]$, where k is the number of mentions. Entities also have their own description, which is obtained from the Wikipedia, $\mathcal{D} = [e_d^1, e_d^2, \dots, e_d^{N'}]$, where $\mathcal{D}$ is the entity descriptions of all entities in the D, and e_d^i is the description of i_{th} entity.

4.2. Framework

The goal of our proposed CPT-RI is to better elicit knowledge from PLMs via prompt tuning and empower with reasoning ability for document RE. As shown in Fig. 2, we first define the Generic Prompt Template (GP) that contains the document, template, and entity description for all entity pairs. Second, BERT is used to encode the GP to capture the hidden states. Then, to enhance factual knowledge for each entity pair, we employ the Relation-specific Prompt Update Module (RPU) to update all information (e.g., entities, and relation information). Next, to use and model the high-order reasoning information for context and relation, we design the Relation Inference Module (RI). Finally, we use the classification module to predict the relations of entity pairs.

4.2.1. Collective Prompt Tuning

Collective Prompt Tuning mainly contains two modules of Generic Prompt Template Module (GP) and Relation-specific Prompt Update Module (RPU): (1) GP aims at eliciting high-level knowledge that is shared across all entity pairs. Thus, it is a common input format for PLMs. (2) RPU is to further elicit relation-specific knowledge for each entity pair. It takes the output of PLMs and re-forms them as soft prompts for multi-head attention. Note that each pair of entities refers to a different prompt.

4.2.2. Relation inference

Relation Inference Module (RI) aims to model high-order relations among relations for better reasoning ability. RI contains three sub-modules: (1) Semantic segmentation module that aggregates related relations to reinforce each other. (2) Criss-cross Constraints for relation matrix to limit the entity interactions to mitigate the noisy issue. (3) Classification module that takes both the outputs of prompt tuning and RI for final relation prediction.

The notations and symbols frequently used in this paper are displayed in Table 1.

5. Collective prompt tuning module

5.1. Generic prompt template

To address the complex and overlong prompt issue, this module is to capture the common parts among each specific prompts, and use entity/relation placeholders for further update.

Prompt Design. We define the generic prompt as $S = [D, T, \mathcal{D}]$, where D denotes the document, $\mathcal{D}$ denotes entity description, which we will describe later, and T is the template for entity pair. Here, we define it as follows: T = `The relation of` [e^s_{mask}] `and` [e^o_{mask}] `is` [r_{mask}] `.`, where e^s_{mask}, e^o_{mask}, and r_{mask} are head entity, tail entity, and relation placeholders, respectively. At the initialization of the model, these placeholders are completely identical.

The generic prompt has two advantages: (1) When faced with hundreds of entity pairs in the document, a generic template can conduct the representation of many relational templates without much manual design, compared to traditional relationship

Table 1
Notations in our model.

Name	Description
S	Generic prompt
D	Document
$\mathcal{D}$	All entity description
T	Template for entity pairs
e^s_{mask}	Placeholder of head entity
e^o_{mask}	Placeholder of tail entity
r_{mask}	Placeholder of relation
CLS	First token of the document
S_{sub}	Generic prompt for the entity pair
D_{sub}	Document information for the entity pair
T_{sub}	Updated generic template for the entity pair
h^d_s	Description of the head entity
h^d_o	Description of the tail entity
h^e_s	Hidden state of the head entity
h^e_o	Hidden state of the tail entity
h_c	Context information of the entity pair
h_r	Updated r_{mask}
H_{sub}	Hidden states of sub-sequence for the entity pairs
G	Reason Mask

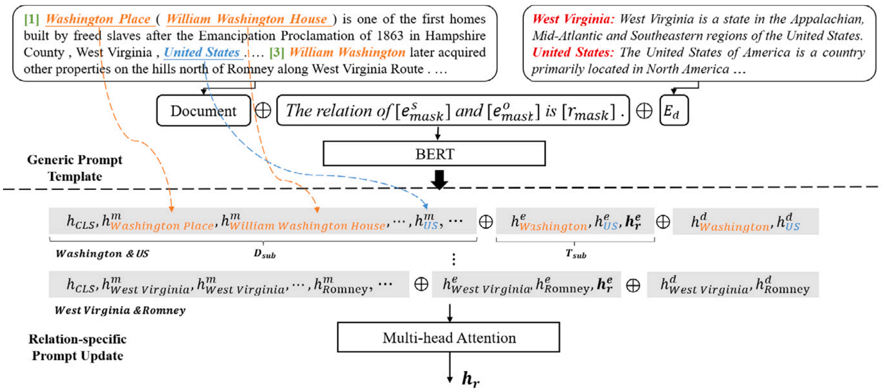

Fig. 3. Collective Prompt Tuning Module. h^m, h^e, and h^d are mention embedding, entity embedding, and entity description embedding, respectively.

extraction. (2) Generic templates can avoid too-long inputs for the encoder. It is not realistic to represent each entity pair as a template, due to the excessive number of entity pairs. In this way, we do not need exhaustive prompts according to the number of entity pairs, but a single one as the input of PLMs.

Entity Description. Entity description can provide explanations and is demonstrated helpful for many knowledge-driven tasks (Basu, Varanasi, Shakerin, Arias, & Gupta, 2021; Guo, He, Dang, & Wang, 2020). We then extract them from Wikipedia,[1] which denotes $\mathcal{D}$. Note that we only extract the first 10 words of the entity in Wikipedia as the entity description and fill them in the tail of the document, which can avoid the overlong input. Another advantage of the entity description is that it has the possibility to contain information about the entity type. Using the entity description is equivalent to turning the entity type information into a sequence of sentence information.

Encoder. BERT (Devlin et al., 2019) is a widely used PLM. We choose BERT$_{base}$ here following most RE baselines for fair comparisons, though our proposed framework is flexible to other PLMs. When the length exceeds the length limit, we take the first 512 tokens. Note that only a few cases have overlong inputs using our collective prompt tuning. We encode the general prompt template $S = [D, T, \mathcal{D}]$ as follows:

$$\mathbf{H}, \mathbf{CLS} = \mathrm{BERT}(S), \tag{1}$$

where $\mathbf{H} = [\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_n]$ is considered as document-level features, and each $\boldsymbol{h}_i$ is a vector that denotes the token features. **CLS** is the hidden state of document feature at the first token of the last layer of the BERT.

[1] www.wikipedia.org.

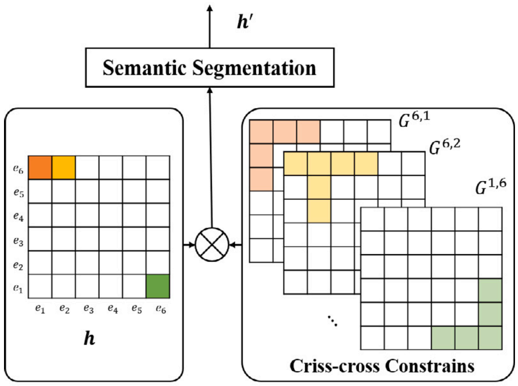

Fig. 4. Interactive entity-based Relation Inference. In the left figure, each box represents the relation information of different entity pairs. In the right figure, each matrix represents the candidate range of target relation information by Criss-cross Constraints.

5.2. Relation-specific Prompt Update

Relation-specific Prompt Update Module (RPU) aims to use different templates to prompt factual knowledge for each entity pair. It takes the encoded generic prompts as inputs, updates entity/relation placeholders, reuses and re-organizes the embeddings to form different prompt templates. Such that each prompt template contains sufficient relation-specific knowledge for the corresponding entity pair.

As shown in Fig. 3, for `Washington` and `US`, we define its relation-specific prompt as follows:

$$\mathbf{S}_{sub} = [\mathbf{D}_{sub}, \mathbf{T}_{sub}, \boldsymbol{h}^d_s, \boldsymbol{h}^d_o], \tag{2}$$

where $\mathbf{D}_{sub} = [\boldsymbol{h}_{CLS}, \boldsymbol{h}^m_{Washington}, \dots, \boldsymbol{h}^m_{US}]$ is the context information, and $\boldsymbol{h}^d_s$ and $\boldsymbol{h}^d_o$ are the entity description embeddings about the entity pair. The prefix m in the $\mathbf{D}_{sub}$ indicates the mention of entity. $\mathbf{T}_{sub} \in \mathbb{R}^{3\times K}$ is the updated generic template by adding the entity placeholder ($\boldsymbol{h}^e_s \in \mathbb{R}^{1\times K}$ and $\boldsymbol{h}^e_o \in \mathbb{R}^{1\times K}$) with entity embeddings ($\boldsymbol{z}_{e^s}$, $\boldsymbol{z}_{e^o}$), which can be computed as follows:

$$\begin{cases} \mathbf{T}_{sub} = [\boldsymbol{h}^e_s + \boldsymbol{z}_{e^s}, \boldsymbol{h}^e_o + \boldsymbol{z}_{e^s}, \boldsymbol{h}^e_r] \\ \boldsymbol{z}_{e^s} = \tanh(\boldsymbol{h}_s \mathbf{W}_s + \hat{\boldsymbol{h}}_c \mathbf{W}_{c1}) \\ \boldsymbol{z}_{e^o} = \tanh(\boldsymbol{h}_o \mathbf{W}_o + \hat{\boldsymbol{h}}_c \mathbf{W}_{c2}) \end{cases}, \tag{3}$$

where $\mathbf{W}_s \in \mathbb{R}^{K\times K}$, $\mathbf{W}_o \in \mathbb{R}^{K\times K}$, $\mathbf{W}_{c1} \in \mathbb{R}^{K\times K}$, and $\mathbf{W}_{c2} \in \mathbb{R}^{K\times K}$ are the trainable parameters. $\boldsymbol{h}^e_r \in \mathbb{R}^{1\times K}$ is the r_{mask} feature in the $\mathbf{H}$. $\boldsymbol{h}_s \in \mathbb{R}^{1\times K}$ and $\boldsymbol{h}_o \in \mathbb{R}^{1\times K}$ are the textual representations of entities by computing the average vectors of all of the corresponding mention embeddings in the D_{sub}. $\hat{\boldsymbol{h}}_c \in \mathbb{R}^{1\times K}$ is a context matrix, where each element denotes entity-to-entity relevance. K is the embedding dimension. Concretely, we use context-based strategy (Zhang, Chen et al., 2021) to obtain $\boldsymbol{h}_c$ as follows:

$$\begin{cases} \boldsymbol{h}_c = \mathbf{W}_c \mathbf{H} \boldsymbol{a}^{(s,o)} \\ \boldsymbol{a}^{(s,o)} = \mathrm{Softmax}(\sum_{i=1}^{v} \mathbf{A}^s_i \cdot \mathbf{A}^o_i) \end{cases}, \tag{4}$$

where $\mathbf{W}_c$ is trainable parameters. $\mathbf{A}^s_i$ and $\mathbf{A}^o_i$ are the attention weight of the entity pair in the i_{th} heads of BERT. v is the number of head in transformer.

Finally, we re-encode the relation-specific template to capture the updated contextual information:

$$\mathbf{H}_{sub} = \text{Mul-ATT}(\mathbf{S}_{sub}), \tag{5}$$

where $\mathbf{H}_{sub} \in \mathbb{R}^{1\times M\times K}$ is the hidden state of sub-sequence. M is the length of $\mathbf{S}_{sub}$. Then, we can obtain the updated feature of r_{mask}, $\hat{\boldsymbol{h}}_r \in \mathbb{R}^{1\times K}$, from the $\mathbf{H}_{sub}$. Finally, we concatenate relation features as $\boldsymbol{h}_r \in \mathbb{R}^{N\times K}$ and concatenate contextual features as $\boldsymbol{h}_c \in \mathbb{R}^{N\times K}$ from the entire document. N is the number of entity pairs in the document.

6. Relation inference module

Relation Inference Module (RI) aims to learn the interaction of relations using the reason graph matrix. RI contains three modules: Semantic Segmentation, Criss-cross Constraints for relation matrix, and Classification.

6.1. Semantic segmentation

We explore the semantic segmentation (Zhang, Chen et al., 2021) to reason the target relation by some important candidate relations. We change the relation matrices of entity pairs to a pixel, which is more like an image such as the left part of Fig. 4. Clearly, considering all relations as candidate information of the target relation will be introduced many non-target relation information, such as negative label information in Fig. 1. Therefore, we use criss-cross constrains mask, $\mathbf{G}^{s,o}$ (Detailed in Section 6.2), to gather related relations for reasoning.

Specially, we transform the size of relation matrix ($\boldsymbol{h}_r$ or $\boldsymbol{h}_c$) to $N' \times N' \times K$. Given a entity pair (e^s and e^o), we can obtain the mask $\mathbf{G}^{s,o}$ and the new relation matrix. Then, we use the popular semantic segmentation model, Self-attention (SA) (Vaswani et al., 2017; Zhang, Goodfellow, Metaxas, & Odena, 2019), to capture the entity-level relation matrix, which can be computed as follows:

$$\begin{cases} att_{i,j} = \frac{(\boldsymbol{h}_i \mathbf{W}_i)(\boldsymbol{h}_j \mathbf{W}_j)^T}{\sqrt{d}} * G^{s,o}_{i,j} \\ \boldsymbol{\alpha} = \mathrm{Softmax}(\mathbf{att}) \\ \boldsymbol{h}' = \boldsymbol{\alpha} \boldsymbol{h} \end{cases}, \tag{6}$$

where $\boldsymbol{h}$ is the relation matrix for the entity pair, and d is the dimension of hidden. $\mathbf{W}_i \in \mathbb{R}^{K \times K}$ and $\mathbf{W}_j \in \mathbb{R}^{K \times K}$ are trainable parameters.

To infer the target relation through other crucial relations, the RI is proposed and used in Eq. (3) to modify the original relation matrices, $\hat{\boldsymbol{h}}_r$ and $\hat{\boldsymbol{h}}_c$, to new matrices, $\hat{\boldsymbol{h}}'_r$ and $\hat{\boldsymbol{h}}'_c$. In specific, we use the semantic segmentation to aggregate the related relation matrices and obtain the new relation matrices, $\hat{\boldsymbol{h}}'_c$. Then, $\hat{\boldsymbol{h}}'_c$ is introduced into Eq. (3) in place of the original $\hat{\boldsymbol{h}}_c$, and get the new updated generic template, $\boldsymbol{T}_{sub}$. Next, the relational representation of $\hat{\boldsymbol{h}}'_r$ is impacted by $\boldsymbol{T}_{sub}$ (Eq. (2)). Finally, the new relational matrices ($\boldsymbol{z}_{e^s}$, $\boldsymbol{z}_{e^o}$, and $\hat{\boldsymbol{h}}'_r$) are applied to predict the relation in Eq. (8).

6.2. Criss-cross constraints for relation matrix

Inspired by Huang et al. (2019), we propose criss-cross constraints to build a mask for semantic segmentation $\mathbf{G} \in \mathbb{R}^{N' \times N'}$, where N' is the number of entities, as shown in the right part of Fig. 4 (Criss-cross Constrains). Each mask, $\mathbf{G}$, denotes the important candidate information for the information of entity pair (e_s and e_o). The $\mathbf{G}$ is defined as follows:

$$G_{i,j} = \begin{cases} 1 & \begin{matrix} if\ (i = s\ or\ j = o) \\ \&|(i+j)-(s+o)| < num \end{matrix} \\ 0 & otherwise \end{cases}, \tag{7}$$

where i, j are the index of row and column about the entity pair. *num* is the distance from the target node to the current node.

The underlying hypothesis behind $\mathbf{G}$ is that: **(1)** Two pairs of entities share at least one entity. As shown in Fig. 1, (`Ralph`, r_1, `US`) and (`William`, r_2 `US`) share a common entity (`US`), thus build a path ($r_1 \leftrightarrow r_2$) for inference. Otherwise, (`Ralph`, r_1, `US`) and (`West`, r_3, `Hampshire`) will never have interactions ($r_1 \nleftrightarrow r_3$). **(2)** The distance of entities' index for two pairs of entities does not exceed a certain threshold (*num*). For example, i, j, s, and o are indexes for two entity pairs in the Eq. (7). The reason is that when the distance of index is too far, the relation categories are different with high probability. We thus use cross-criss constraints to remove noisy interactions from semantic segmentation.

6.3. Classification module

For entity pair (e^s, e^o), we use bilinear function to obtain the $\boldsymbol{h}_{e,o}$, and add the $\hat{\boldsymbol{h}}'_r$ as the relation matrix. Then, we obtain the probability of being target relation by a linear function, which can computed as follows:

$$\begin{cases} \boldsymbol{h}_{e,o} = \tanh(\boldsymbol{z}_{e^s} \mathbf{W}_t \boldsymbol{z}_{e^o} + b_t) \\ P(r|e^s, e^o) = (\boldsymbol{h}_{e,o} + \hat{\boldsymbol{h}}'_r)\mathbf{W}_r + b_r \end{cases}, \tag{8}$$

where $\mathbf{W}_t$, $\mathbf{W}_r$, b_t, and b_r are the trainable parameters. Finally, we use Adaptive Thresholding (Zhou et al., 2021) to estimate the model parameters and obtain the labels of entity pairs. In this paper, we employ dropout to prevent overfitting.

7. Experiment

We will first describe the datasets and experimental details, then give the experimental results and analysis. We also conduct the case study.

7.1. Datasets and metrics

To demonstrate the performance of our model, we evaluate the model on the DocRED[2] (Yao et al., 2019), CDR,[3] and GDA[4] (Wu, Luo, Leung, Ting, & Lam, 2019).

DocRED. The document-level RE dataset (DocRED) annotates named entities and relations. DocRED (human-annotated) contains 132,375 entities, and 56,354 relational facts annotated on 5053 Wikipedia document. And there are 96 relation types which cover a broad range of categories.

CDR. The Biocreative V Chemical Disease Relation benchmark (CDR) (Li et al., 2016) was derived from the Comparative Toxicogenomics Database (CTD), which curates interactions between genes, chemicals, and diseases. And CDR contains 1500 documents with 4409 annotated chemicals, 5818 diseases, and 3116 chemical-disease interactions.

GDA. The Gene Disease Associations (GDA) dataset (Wu et al., 2019) is a RE dataset in the biomedical domain, which contains a relation between gene and disease: Gene-Induced-Disease. And GDA contains 30,192 documents and is split to three parts: train dataset (23,353), dev dataset (5839), and test dataset (1000).

Following the previous work (Yao et al., 2019; Zeng et al., 2020), we utilize the F1 and Ign F1 as the evaluation metrics to evaluate the performance of methods. Ign F1 is calculated excluding relational facts shared by the Train and Dev/Test sets. We also use the intra-F1 and inter-F1 metrics to evaluate a model's performance on intra-sentence relations and inter-sentence relations on the Dev.

7.2. Parameter settings

Our main goal is to verify our proposed collective prompt tuning and the integration with reasoning module. Thus, we choose the most widely used BERT-base as basic PLMs, to avoid exhaustive parameter tuning. The learning rate is set to $1e{-}5$ for pre-trained model and $1e{-}4$ for others. We conduct grid search to tune hyperparameters: the size of embedding is in $\{64; 128; 256; 512; \mathbf{768}\}$, where bold font denotes the best setup. The batch size for pre-training is set to 2. The dropout rate is set to 0.2. The constraint distance of the reason graph is 30. And we use 4 layers of semantic segmentation and Mul-ATT. We use cased BERT-base (Devlin et al., 2019) or RoBERTa-large (Liu et al., 2019) as the encoder on the DocRED and utilize cased SciBERT-base (Beltagy, Lo, & Cohan, 2019) as the encoder on the CDR and GDA. We train our model with 2080Ti.

7.3. Baseline

We compare the CPT-RI with the following baselines.

- **Sequence-based methods.** These models use different neural architectures to encode entire document, include CNN, LSTM, Bi-LSTM, Context-Aware, LSR-Glove (Nan et al., 2020), and HIN-GloVe (Tang et al., 2020).
- **Graph-based Models.** These models construct mention/entity graphs, including GAT (Velickovic et al., 2018), BRAN (Verga, Strubell, & McCallum, 2018), GCNN (Sahu, Christopoulou, Miwa, & Ananiadou, 2019), EoG (Christopoulou, Miwa, & Ananiadou, 2019), and AGGCN (Guo, Zhang, & Lu, 2019).
- **Transformer-based Models.** These models utilize pre-trained language model to predict relations, include BERT-Two-Step (Wang, Focke, Sylvester, Mishra, & Wang, 2019), BERT-RE (Wang et al., 2019), HIN-BERTb (Tang et al., 2020), LSR-BERT (Nan et al., 2020), CorefBERT (Ye et al., 2020), ATLOP (Zhou et al., 2021), GAIN (Zeng et al., 2020), MRN (Li, Xu et al., 2021), DocuNET (Zhang, Chen et al., 2021), and KD-DocRED (Tan et al., 2022).

7.4. Overall performance

We show CPT-RI's performance[5] on the DocRED,[6] CDR and GDA in Tables 2 and 3, in comparison with other baselines.

We can see that:

(1) CPT-RI consistently outperforms all baselines on both datasets. This is mainly because we can better utilize PLMs via prompt tuning and empowered with reasoning ability.

(2) In Table 2, the models with BERT as encoders yields a great improvement upon those models without BERT. This demonstrates the importance of PLMs and we should pay more research attentions to elicit learnt knowledge.

(3) In Table 3, CPT-RI also outperforms all baselines, even with other PLM, SciBERT-base. Compared with ATLOP and DocuNET, CPT-RI$_{Base}$ improves F1 by 5.70% and -4.90% on the CDR. In addition, we also can observe that our model improves the F1 score by 2.7% and 0.5% on the GDA. The results demonstrate that our method can generalize well to other PLMs.

[2] https://github.com/thunlp/DocRED.

[3] http://www.biocreative.org/.

[4] https://github.com/geomdata/gda-public.

[5] Note that, the most important baseline is ATLOP. To integrate PLMs with reasoning ability and deal with complex and overlong input templates, our code is done on the basis of ATLOP.

[6] Original papers are without reporting the Intra F1 and Inter F1. We run these methods (ATLOP, DocuNET, and KD-DocRED) more than 10 times and take the best F1 among them as the result of comparison experiments.

Table 2
Results on the Dev and Test of DocRED. We report the mean and standard deviation of F1 on the development set by conducting 3 runs of training. Results with † are computed based on re-trained models. Other results are reported in their original papers. The second-best results are underlined.

Model (%)	Dev				Test	
	Ign F1	Intra F1	F1	Inter F1	Ign F1	F1
Sequence-based models						
CNN	41.58	51.87	43.45	37.58	40.33	42.26
LSTM	48.44	–	50.68	–	47.71	50.07
Bi-LSTM	48.87	57.05	50.94	50.94	48.78	51.06
Context-aware	48.94	–	51.09	–	48.40	50.70
HIN_{Glove}	51.06	60.83	52.95	48.35	51.15	53.30
Graph-based models						
GAT	45.17	58.14	51.44	43.94	47.36	49.51
GCNN	46.22	57.78	51.52	44.11	49.59	51.62
EoG	45.94	58.90	52.15	44.60	49.48	51.82
AGGCN	46.29	58.76	52.47	45.45	48.89	51.45
LSR_{Glove}	48.82	60.83	55.17	48.35	52.15	54.18
GAIN-BERT_{Glove}	53.05	61.67	55.29	48.77	52.66	55.08
Transformer-based models						
BERT-RE_{Base}	–	61.61	54.16	47.15	–	53.20
RoBERTa_{Base}	53.85	–	56.05	–	53.52	55.77
$\text{BERT-Two-Step}_{Base}$	–	61.80	54.42	47.28	–	53.92
HIN-BERT_{Base}	54.29	–	56.31	–	53.70	55.60
CorefBERT_{Base}	55.32	–	57.51	–	54.54	56.96
LSR-BERT_{Base}	52.43	65.26	59.00	52.05	56.97	59.05
ATLOP-BERT_{Base}†	59.47	67.16	61.13	52.71	59.31	61.30
GAIN-BERT_{Base}	59.14	67.10	61.22	53.90	59.00	61.24
MRN-BERT_{Base}	59.74	67.74	61.61	54.43	59.52	61.74
$\text{DocuNET-BERT}_{Base}$†	59.26	**68.23**	61.87	54.73	59.81	61.69
$\text{KD-DocRED-BERT}_{Base}$†	59.84	67.94	61.86	54.83	59.23	**61.94**
CPT-RI_{Base}	**60.02 ± 0.18**	<u>68.12 ± 0.21</u>	**62.13 ± 0.15**	**54.95 ± 0.15**	**59.92**	<u>61.87</u>

Table 3
Results on the Test of CDR and GDA. The second-best results are underlined.

Model (F1%)	CDR	GDA
CNN	52.60	–
BRAN	62.10	–
EoG	63.60	81.50
LSR_{Base}	64.80	82.20
$\text{SciBERT-ATLOP}_{Base}$	69.40	83.30
$\text{SciBERT-MRN}_{Base}$	65.90	82.90
$\text{SciBERT-DocuNET}_{Base}$†	**76.50**	<u>85.10</u>
$\text{SciBERT-CPT-RI}_{Base}$	<u>71.60 ± 1.10</u>	**85.60 ± 0.40**

7.5. Detailed analysis

To further analyze CPT-RI, we also conduct ablation analysis to illustrate the effectiveness of different modules and mechanisms in CPT-RI. We show the results of the ablation study in Table 4 and Fig. 5. In the table, CPT-RI_{Base} has a more excellent performance on the DocRED. The reason is that our model uses collective prompt tuning to learn more information about the entities and employs criss-cross constraints to infer the target relation by related relations. These ablation results demonstrate the effectiveness of both Collective Prompt tuning (CPT) and Relation Inference (RI).

Collective Prompt Tuning. We examined the performance of our main components in Table 4. We find that: **(1)** The performance becomes sharply poor without collective prompt tuning. Specifically, the experimental results are reduced by 1.27% F1 and 1.09% Ign F1 on the Dev setting in CPT-RI_{Base}. It isn't easy to extract the relations from the document. Because prompt tuning can bridge the gap between PLMs and RE task, and facilitate the advantages of PLMs. **(2)** When removed general template or entity description, the experimental results are reduced. There are two reasons. On the one hand, the general template can learn relational information and is helpful for RE. On the other hand, entity description can introduce rich external knowledge and increase the degree of distinction among relations. **(3)** Without the relation-specific prompt update, the drop of performance indicates that only generic prompt is not enough. The specific prompts can further elicit knowledge for each entity pair efficiently.

Relation Inference. In Table 4, we can also observe the following insights for relation inference module: **(1)** Without relation inference, the experimental results drop, indicating that it is rational and necessary to integrate complementary PLMs and reasoning module. **(2)** When we remove semantic segmentation, the F1 scores drop. The reason is twofold. On the one hand, the relations of entity pairs are very complicated, such as long-distance between entity pairs, similar context, relational nesting, etc. **(3)** Without the

Table 4
An ablation study for the CPT-RI model.

Model (%)	Dev	
	Ign F1	F1
CPT-RI$_{Base}$	**60.02**	**62.13**
w/o Collective prompt tuning	58.93	60.86
w/o Generic template	59.31	61.19
w/o Entity description	59.25	61.37
w/o Relation-specific prompt update	59.51	61.24
w/o Relation inference	59.28	61.20
w/o Semantic segmentation	59.51	61.47
w/o Criss-cross constraints	59.16	61.43

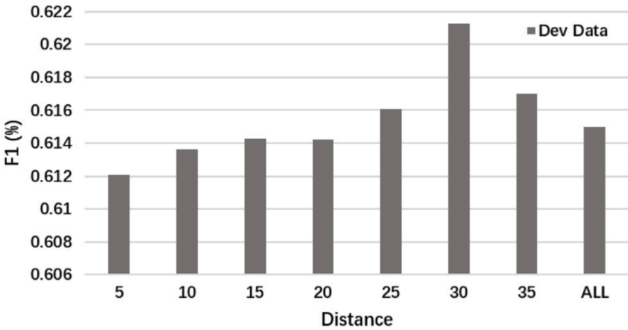

Fig. 5. Impacts of the length of reasoning path on the Dev (DocRED).

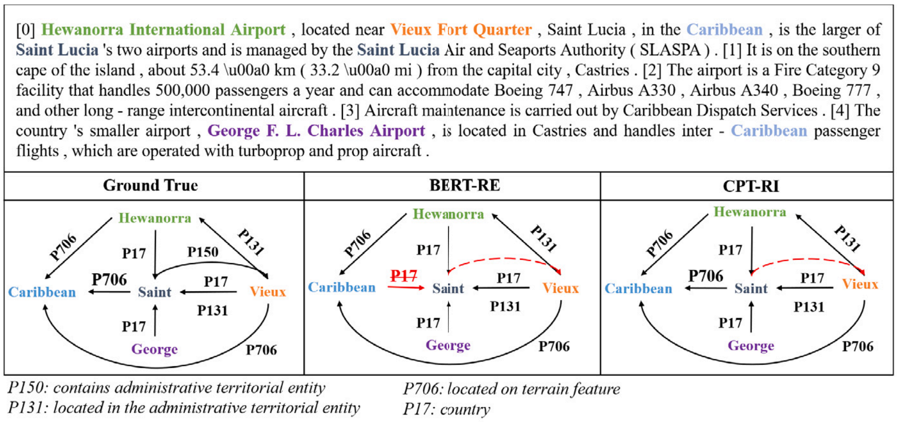

Fig. 6. The case study of our proposed CPT-RI and BERT-RE models. The red line indicates an incorrect relation or relation between the entity pair is not recognized. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

criss-cross constraints, the results of the experiment are reduced by 0.70% F1 and 0.86% Ign F1 on Dev in CPT-RI$_{Base}$. The major reason is that the criss-cross constraints can alleviate the noisy information and capture the appropriate candidate relations. On the other hand, semantic segmentation can capture the dependency of relations in the document.

In general, we can observe that the experimental results drop when the CPT or RI modules are removed. And the experiment works best when CPT and RI modules are applied. This demonstrates that these two modules have no inhibitory effect. Therefore, it is proved that these two modules have a positive impact on the model and can improve each other.

Constraints Distance. In Fig. 5, We investigate the effect of distance *num* from the target node to the current node. As the constraints distance increases, the F1 improves. When the *num* reaches 30, the F1 achieves the best. Then, the performance of F1 drops. The reason is that the increase of constraint distance means that the target relation and non-target relations increase simultaneously. However, non-target relations tend to grow faster than the helpful relation in the document-level RE dataset. Therefore, an appropriate amount of relations needs to be introduced to avoid a drastic increase in non-target relations.

Table 5
An ablation study for the CPT-RI model.

Model (%)	Dev	
	Ign F1	F1
T (our method)	**60.02**	**62.13**
T_1	59.81	61.73
T_2	58.58	61.62
T_3	59.33	61.15

7.6. Effect of template

To further analyze the effect of template on the document-level RE, we design three extra templates: "T_1 = `In summary, [`e^s_{mask}`] and [`e^o_{mask}`] are [`r_{mask}`].`", "T_2 = `[`e^s_{mask}`] and [`e^o_{mask}`] are [`r_{mask}`].`", and "T_3 = `[Their relation is [`r_{mask}`].`". The template of our method is "T = `The relation of [`e^s_{mask}`] and [`e^o_{mask}`] is [`r_{mask}`].`". We show the results in Table 5.

We can find that in Table 5: (1) Our template (T) achieves the best performance than other templates. (2) The T_3 is the worst result. The reason is that this template (T_3) lacks the information of the entity pair. (3) The T_1 and T_2 have the similar results. And they are worse than T, but better than T_3. This demonstrates two things: first, the placeholder of entity information is very important. Second, the important cue words (e.g., relation) can improve performance. (4) The different templates impact the relations of entity pairs and are more complex than they may seem. Numerous factors influence the results of the experiment.

7.7. Case study

As shown in Fig. 6, we present a case study to give intuitive impression of our proposed CPT-RI, in comparison with BERT-RE. We can observe that `Vieux Fort Quarter` is the subject of the triples: (`Vieux Fort Quarter`, located in the administrative territorial entity, `Saint Luica`) and (`Vieux Fort Quarter`, located in the administrative territorial entity, `Hewanorra International Airport`). However, BERT-RE fails to identify the relation between `Saint Luica` and `Caribbean`, while CPT-RI deduces it successfully. BERT-RE and CPT-RI can predict the right label for (`Hewanorra`, `Caribbean`), but for (`Saint Luica`, `Caribbean`) BERT-RE predicts the wrong label. These two entity pairs share the entity (`Caribbean`), indicating that CPT-RI can capture more useful information about relation and strong capability for reasoning. Meanwhile, BERT-RE and CPT-RI do not recognize that there is a relation between `Saint Luica` and `Vieux Fort Quarter`. This indicates it is a difficult point to study the relations between the entity pair in which subject and object are reversed.

8. Conclusions

In this paper, we exploit a novel Collective Prompt Tuning with Relation Inference (CPT-RI) for Document-level RE Model, which can efficiently utilize prompt tuning for complex tasks and empower with reasoning ability. Experimental results show that our method is effective and significantly better than competitive baselines. The extensive analysis confirms that the collective prompt tuning can better elicit useful knowledge from generic to specific levels from PLMs. Meanwhile, the reasoning module is also necessary to further boosts performance by modeling relations among relations.

In the future, we are interested in applying CPT-RI to more complex tasks and exploring more advanced reasoning modules.

CRediT authorship contribution statement

Changsen Yuan: Methodology, Software, Coding, Writing – original draft. **Yixin Cao:** Supervision, Writing – review & editing. **Heyan Huang:** Supervision, Writing – review & editing.

Data availability

I have provided the dataset link in the paper.

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